DESIGN FOR TESTABILITY (DFT) AND PHYSICAL VERIFICATION (LVS/DRC)

INTRODUCTION

Objective:

The purpose of this task is to understand and implement Design-for-Testability (DFT) using scan chains and to learn basic physical verification techniques such as Layout vs Schematic (LVS) and Design Rule Check (DRC).

Scope:

1. DFT / Scan Insertion: Concept, manual approach, test vectors, and fault coverage.
2. Basic Physical Verification: DRC and LVS concepts, workflow, and expected outcomes.

DFT / Scan Insertion

Design-for-Testability (DFT) is a technique used to make digital circuits easier to test for manufacturing faults. It improves the ability to detect faults efficiently after fabrication.

One of the most commonly used DFT techniques is **scan insertion**, where flip-flops are connected in a series chain known as a **scan chain**.

Each flip-flop operates in two modes:

- **Normal Mode:** The circuit performs its intended functionality.
- **Scan Mode:** Flip-flops are connected as a shift register for testing purposes.

SCAN SIGNALS:

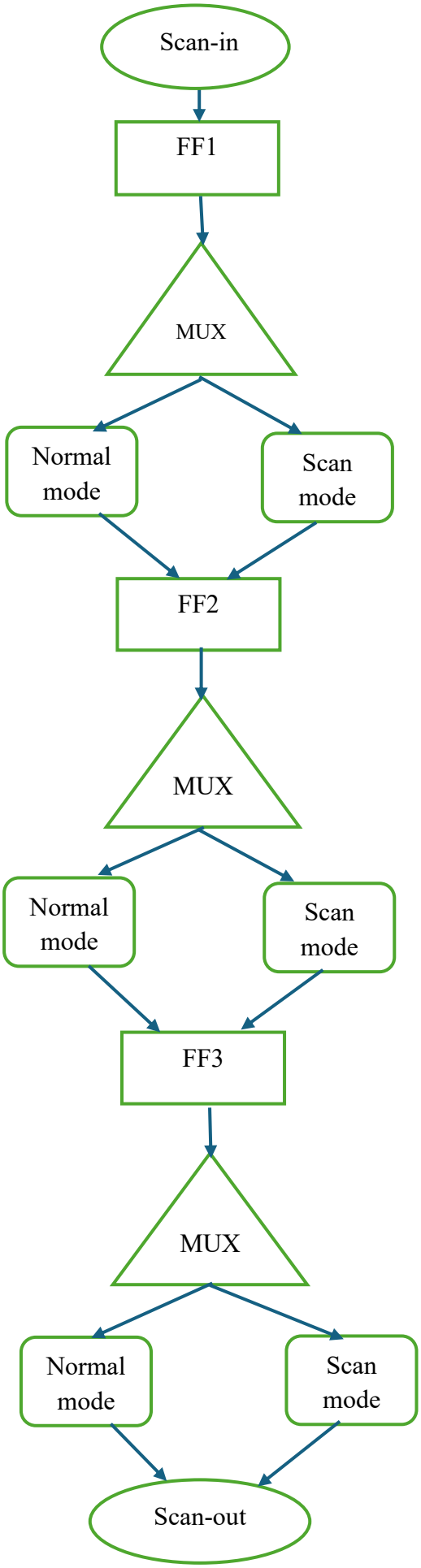
Scan in: Input for shifting test data

Scan out: Output for observing test response

Scan enable: Selects scan mode

Clock: Used to shift data

Diagram:


Scan-in
FF1
MUX
Normal mode
Scan mode
FF2
MUX
Normal mode
Scan mode
FF3
MUX
Normal mode
Scan mode
Scan-out

Explanation Of Diagram:

The above diagram represents a scan chain consisting of multiple flip-flops connected in series. A multiplexer is used at each flip-flop to select between normal mode and scan mode. Test data is shifted from Scan-In through the flip-flops and observed at Scan-Out for fault detection.

Test Vector Table:

Test Pattern	Fault Detected
101	Detects SA0 at FF2
110	Detects SA1 at FF1
011	Detects SA0 at FF3

Stuck-at Faults:

A stuck-at fault occurs when a signal is permanently stuck at logic ‘0’ or logic ‘1’. These faults can affect the normal operation of the circuit and must be detected during testing.

Test Coverage:

Test coverage refers to the percentage of faults detected by the applied test patterns. Scan chains improve test coverage by providing better access to internal flip-flops, allowing most stuck-at faults to be detected efficiently. Test Coverage formula:

Test Coverage = Detected Faults / Total Faults × 100

Example:

Detected = 90

Total = 100

Coverage = 90%

Basic Physical Verification (LVS/DRC):

1. Introduction to Physical Verification

Physical verification ensures that the layout of a circuit is correct and manufacturable. It checks whether the design follows fabrication rules and matches the original schematic.

2. Design Rule Check (DRC)

Design Rule Check (DRC) is used to verify that the layout follows the manufacturing rules defined by the fabrication process. These rules include constraints on spacing, width, and alignment of different layers. DRC helps identify violations that may cause issues during fabrication.

3. Layout vs Schematic (LVS)

Layout vs Schematic (LVS) is used to compare the layout design with the original schematic. It ensures that the connectivity and components in the layout match the schematic. LVS helps detect errors such as missing connections or incorrect wiring.

4. Steps Involved in LVS/DRC

The basic steps involved in physical verification are:

- Run DRC to check for layout rule violations
- Fix any DRC errors
- Run LVS to compare layout with schematic
- Fix any mismatches found in LVS
- Verify that the design is clean and ready for fabrication

Result:

- Scan chain successfully inserted conceptually
- Stuck-at faults explained and test vectors generated
- DRC flow studied
- LVS flow studied
- Physical verification steps understood

Conclusion:

In this task, the concepts of Design-for-Testability (DFT) and basic physical verification were studied. Scan chain insertion helps in detecting faults such as stuck-at faults efficiently, improving test coverage.

Additionally, physical verification techniques like DRC and LVS ensure that the layout follows manufacturing rules and correctly represents the schematic. These steps are essential in ensuring a reliable and manufacturable design.